

A Packing of 20 Four-Spaces in $\mathbb{R}^{10}$

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Abstract

We give 20 4-dimensional subspaces of $\mathbb{R}^{10}$ whose smallest squared chordal distance is at least 2.525899. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.447483, obtained by deleting 30 of the 50 four-spaces of the packing listed in Cohn’s current table. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 20 subspaces of dimension 4 in $\mathbb{R}^{10}$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 2.525899 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane’s *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 10$, $n = 4$. The cell $N = 20$ is blank in Sloane’s table, in Sloane’s coordinate files and in Cohn’s table as checked on October 8, 2026. The best value implied by known packings is 2.447483, obtained by deleting 30 of the 50 four-spaces of the packing listed in Cohn’s current table. The Rankin simplex bound for this cell is $4(10 - 4)/10 \cdot 20/19 = 2.526316$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m10n4N20.txt`

in <https://github.com/Ge-limin/math719>.

*Found and checked by Claude, a model released by Anthropic, running on the author’s laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 65 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
631474834684004433185785342092557819510637588881973390359194978296970366
692479001111352514447476618703704500312079786828791172545442827197431930
808559227160949122247960820488089545973495825758
```

denominator:

```
249999999999826303023260446040086517860136408402030320706720495857414088
517756730620259565547993662355551491817877493547482706585125142218071068
175047220121768067876194118507697913944816881103
```

B Coordinates

20 blocks of 10 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m10n4N20.txt`.

```
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